Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

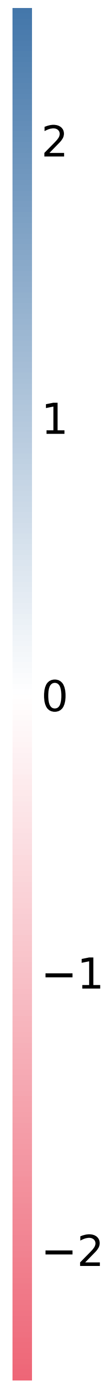
F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV



Reasoning + Answer:

<think>

The de Broglie wavelength (Î») of an electron can be calculated using the formula: Î» = h / âĪļ(2m*K), where h is Planck's constant, m is the mass of the electron, and K is the kinetic energy.

The given kinetic energy is 1 eV = 1.6 ÃĹ 10^(-19) J. Rearrange the formula to solve for Î»: Î» = h / âĪļ(2m*K)

Substitute values, we get Î» = h / âĪļ(2 ÃĹ 9.11 ÃĹ 10^(-31) * 1.6 ÃĹ 10^(-19))

Simplify and convert to unit cm, then the de Broglie wavelength Î» âīĪ 1.23 ÃĹ 10^-7 cm.

</think><answer> G